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**SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY & MANAGEMENT, BANTAKAL**

**Model Questions**

E-Waste Management- BEC755A

**2025-26**

| Qn. No | Question | Marks | BL* | TLO* | CO* |
|---|---|---|---|---|---|
| | **MODULE 3** | | | | |
| **1** | Explain why e-waste is classified as post-consumer waste in the e-waste management framework. | | 2 | 3.1 | 3 |
| **2** | Describe the major stages of the e-waste value chain from generation to recycling. | | 2 | 3.1 | 3 |
| **3** | Differentiate between formal and informal e-waste collection systems with suitable points. | | 2 | 3.1 | 3 |
| **4** | Summarize the role of authorized collection centers in improving the e-waste value chain. | | 2 | 3.1 | 3 |
| **5** | Explain how consumer awareness affects the effectiveness of e-waste collection systems. | | 2 | 3.1 | 3 |
| **6** | Explain the concept of Extended Producer Responsibility (EPR) in the context of e-waste management. | | 2 | 3.2 | 3 |
| **7** | Describe how Collective Responsibility supports effective implementation of EPR. | | 2 | 3.2 | 3 |
| **8** | Explain the role of a Producer Responsibility Organization (PRO) in assisting producers to meet EPR obligations. | | 2 | 3.2 | 3 |
| **9** | Differentiate between Individual Producer Responsibility (IPR) and Collective Producer Responsibility (CPR). | | 2 | 3.2 | 3 |
| **10** | Summarize the key functions performed by PROs in the e-waste collection and recycling value chain. | | 2 | 3.2 | 3 |
| | **MODULE 4** | | | | |
| **1** | Classify e-waste into hazardous and non-hazardous categories for a given device. | | 3 | 4.1 | 4 |
| **2** | Apply characterization steps to identify materials in discarded electronics. | | 3 | 4.1 | 4 |
| **3** | Suggest suitable packaging for safe transport of e-waste. | | 3 | 4.1 | 4 |
| **4** | Prepare a proper label for an e-waste container. | | 3 | 4.1 | 4 |
| **5** | Recommend correct packaging and labelling for mixed e-waste items. | | 3 | 4.1 | 4 |
| **6** | Apply safe transportation guidelines to suggest how e-waste should be moved from a collection centre to a recycler. | | 3 | 4.2 | 4 |
| **7** | Recommend proper storage methods for hazardous e-waste items in a lab. | | 3 | 4.2 | 4 |
| **8** | Identify suitable personal protective equipment (PPE) for workers handling mixed e-waste. | | 3 | 4.2 | 4 |
| **9** | Suggest precautionary steps to prevent accidents during manual e-waste handling. | | 3 | 4.2 | 4 |
| **10** | Propose a safe handling procedure for transporting lithium-ion battery waste. | | 3 | 4.2 | 4 |
| | **MODULE 5** | | | | |

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| **1** | Identify hazardous substances present in a discarded mobile phone using ROHS guidelines. | | 3 | 5.1 | 5 |
| **2** | Apply ROHS Directive 84 restrictions to evaluate whether a given electronic device is compliant. | | 3 | 5.1 | 5 |
| **3** | Suggest design changes to reduce lead or mercury content in electronics as per ROHS standards. | | 3 | 5.1 | 5 |
| **4** | Examine an electronic product label and determine if it meets ROHS compliance requirements in India. | | 3 | 5.1 | 5 |
| **5** | Apply Indian ROHS rules to check whether a small manufacturer's product can be sold in the market. | | 3 | 5.1 | 5 |
| **6** | Apply dismantling steps to separate major components from a discarded desktop CPU. | | 3 | 5.2 | 5 |
| **7** | Suggest a segregation method for separating plastics, metals, and PCBs from mixed e-waste. | | 3 | 5.2 | 5 |
| **8** | Identify the suitable recycling technology (mechanical/chemical/pyro-metallurgical) for recovering copper from cables. | | 3 | 5.2 | 5 |
| **9** | Propose a simple recovery process to extract reusable metals from printed circuit boards (PCBs). | | 3 | 5.2 | 5 |
| **10** | Apply standard recycling operations to design a basic flow for processing mobile phone e-waste | | 3 | 5.2 | 5 |

BL* Bloom's Taxonomy Level; TLO* Topic Level Outcome; CO- Course Outcome